Journal of Volcanology and Geothermal Research, 36 (1989) 37–52
Elsevier Science Publishers B.V., Amsterdam — Printed in The Netherlands

COMPLETE GRAVITY STUDY OF PITON DE LA FOURNAISE VOLCANO, REUNION ISLAND

DOMINIQUE ROUSSET[1], ALAIN LESQUER[1], ALAIN BONNEVILLE[2] and JEAN FRANÇOIS LÉNAT[2]

[1]*Centre Géologique et Géophysique, USTL, 34060 Montpellier cedex, France*
[2]*U.A. C.N.R.S. No. 10, Centre de recherches volcanologiques, 5, Rue Kessler, 63038 Clermont-Ferrand cedex, France*

(Received December 23, 1986; accepted for publication September 30, 1987)

Abstract

Rousset, D., Lesquer, A., Bonneville, A. and Lénat, J.-F., 1989. Complete gravity study of Piton de la Fournaise volcano, Réunion Island. In: A.R. McBirney (Editor), Piton de la Fournaise Volcano, Réunion Island. *J. Volcanol. Geotherm. Res.*, 36: 37–52.

We present an interpretation of all available gravity data on Piton de la Fournaise volcano, including a new set collected in the summit area. Using a linear programming inversion method, we determine the extension of the Grand Brûlé ultramafic body, drilled in 1985, and we propose to interpret it as a frozen magma chamber related to a disappeared volcano. We also present evidence for superficial mass excess under the Enclos Fouqué and Plaine des Sables calderas that we link to the present Piton de la Fournaise volcanism.

The detailed gravity survey of the summit area allows us to emphasize the great importance of dyke complexes associated with the NE and SE rift zones and with the shallow magma chamber.

A negative anomaly in the summit area is interpreted as a relative minimum between the positive anomaly of Grand Brûlé and those of the western part of the Enclos Fouqué and Plaine des Sables calderas.

Introduction

Located in the Western Indian Ocean (21°07′S–55°32′E) Réunion Island forms with Mauritius and Rodriguez Islands the Mascarene archipelago. The Island is entirely volcanic and lies on the oceanic part of the African plate in the Mascarene Basin which is more than 4000 m deep, bounded on the east by the Carlsberg Ridge and on the south by the West Indian Ridge.

Réunion Island is the emerged part of a broad (240×270 km) elliptic edifice which consists of two overlapping volcanoes (Fig. 1). Piton des Neiges (3069 m), the older one, is now quiescent. Piton de la Fournaise (2631 m) is still active on the southeastern flank of Piton des Neiges.

The oldest lavas which can be dated are 2.1 Ma old (McDougall, 1971) but volcanism is certainly much older. Piton des Neiges ended activity approximately 20,000 years ago (Lecointre and Vincent, 1984), by emitting differentiated lavas, while Piton de la Fournaise began to rise 400,000 years ago. The volcanism of both Piton des Neiges and Piton de la Fournaise is characterized by an islandwide N120° rift zone (Chevallier and Bachèlery, 1981; Chevallier and Vatin-Pérignon, 1982) probably related to regional oceanic structures.

0377-0273/89/$03.50 © 1989 Elsevier Science Publishers B.V.

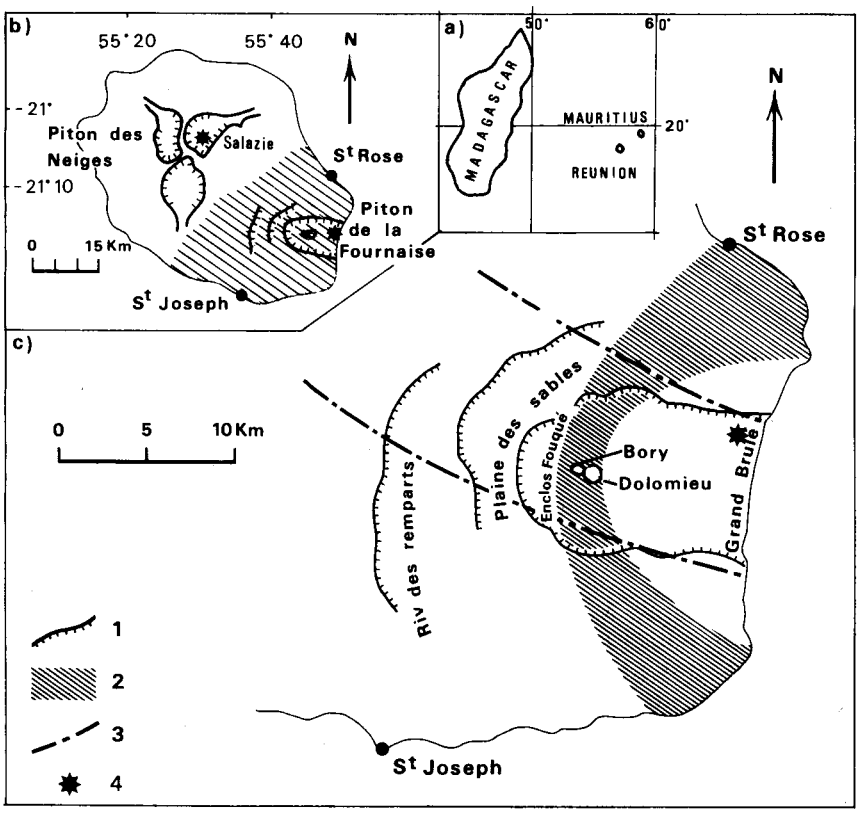

Fig. 1. (a) Réunion Island location in the Indian Ocean; (b) Piton des Neiges and Piton de la Fournaise volcanoes. (c) Major structural features of Piton de la Fournaise volcano. *1* = caldera limits; *2* = NE-SE intrusive riftzone; *3* = supposed N120° rift-zone limits; *4* = drill-holes.

The growth of the Piton de la Fournaise is characterized by the collapse of three roughly concentric and progressively smaller calderas. From west to east, we find (Fig. 1): the Rivière des Remparts caldera, the Plaine des Sables caldera, and the Enclos Fouqué. Presently most of the eruptions occur in the latter. Several authors (Duffield et al., 1982; Lénat and Aubert, 1982) have pointed out the evidence of large eastward-moving landslide features at Piton de la Fournaise.

Geophysical investigations of Piton de la Fournaise volcano are still in progress. A volcanological observatory has been monitoring the dynamic behaviour of the volcano since 1979. Some preliminary gravimetric (Gérard et al., 1980) and magnetic studies (Lénat and Aubert, 1982) have shown that a deep structure is associated with the N120° rift zone. Concerning the northeast- and southeast-trending rift zones, the magnetic survey found dyke complexes where the gravity measurements are too scarcely distributed to yield a definite result. The strong positive gravity anomaly of Grand Brûlé, shown on the 1980 map is now known to close near the shore (Rousset et al., 1987) and, if we refer to the results of the 1985 drill-hole (Rançon et al., 1987), can be attributed to a giant frozen magma chamber whose size and shape are discussed in this paper.

New data, such as a shipborne complete survey of 1984 and a detailed gravity map of the summit zone provided us with the opportunity for a new interpretation of the old gravity data using more sophisticated methods.

In this paper, we present a three-dimensional

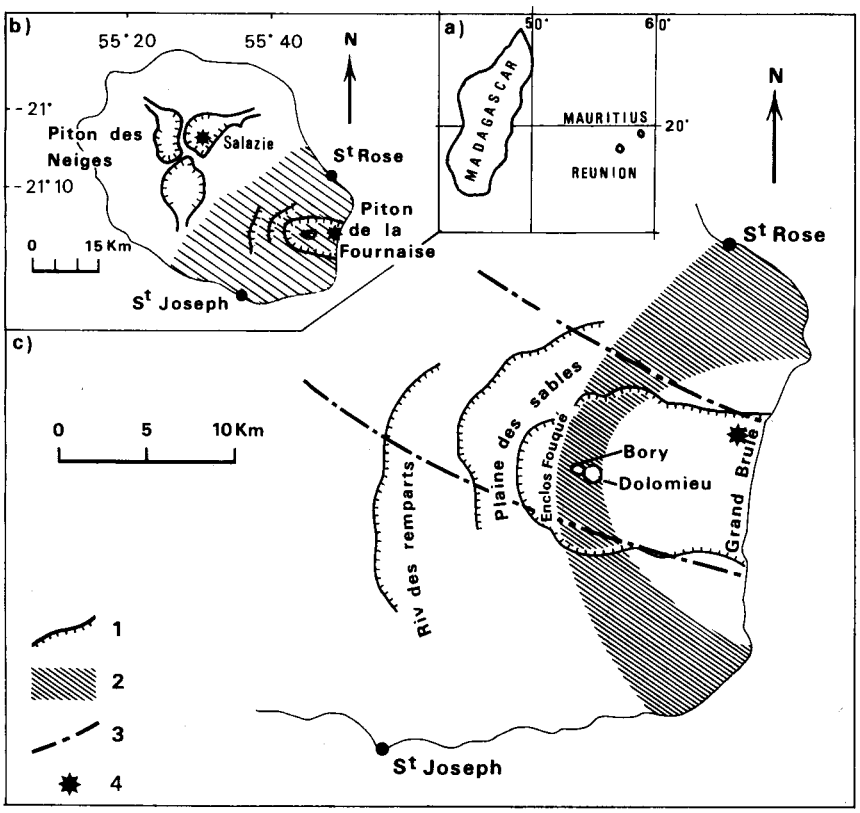

Fig. 1. (a) Réunion Island location in the Indian Ocean; (b) Piton des Neiges and Piton de la Fournaise volcanoes. (c) Major structural features of Piton de la Fournaise volcano. *1* = caldera limits; *2* = NE-SE intrusive riftzone; *3* = supposed N120° rift-zone limits; *4* = drill-holes.

The growth of the Piton de la Fournaise is characterized by the collapse of three roughly concentric and progressively smaller calderas. From west to east, we find (Fig. 1): the Rivière des Remparts caldera, the Plaine des Sables caldera, and the Enclos Fouqué. Presently most of the eruptions occur in the latter. Several authors (Duffield et al., 1982; Lénat and Aubert, 1982) have pointed out the evidence of large eastward-moving landslide features at Piton de la Fournaise.

Geophysical investigations of Piton de la Fournaise volcano are still in progress. A volcanological observatory has been monitoring the dynamic behaviour of the volcano since 1979. Some preliminary gravimetric (Gérard et al., 1980) and magnetic studies (Lénat and Aubert, 1982) have shown that a deep structure is associated with the N120° rift zone. Concerning the northeast- and southeast-trending rift zones, the magnetic survey found dyke complexes where the gravity measurements are too scarcely distributed to yield a definite result. The strong positive gravity anomaly of Grand Brûlé, shown on the 1980 map is now known to close near the shore (Rousset et al., 1987) and, if we refer to the results of the 1985 drill-hole (Rançon et al., 1987), can be attributed to a giant frozen magma chamber whose size and shape are discussed in this paper.

New data, such as a shipborne complete survey of 1984 and a detailed gravity map of the summit zone provided us with the opportunity for a new interpretation of the old gravity data using more sophisticated methods.

In this paper, we present a three-dimensional

inversion of all the available gravity data using linear programming methods.

Gravity data and processing

Data

The data come from various surveys carried out by the Office de la Recherche Scientifique et Technique Outre-Mer, the Bureau de Recherches Géologiques et Minières and the Centre Géologique et Géophysique for the onshore part (Fig. 2). Measurements performed between 1974 and 1979 (450 stations) were used to compute the first Bouguer anomaly map of the southeastern part of Réunion Island (Gérard et al., 1980). The gravity map established after the "Fournaise" campaign, carried out in 1984, (750 values) of the French R/V Jean Charcot (Rousset et al., 1987) is added to give a complete Bouguer anomaly map of Piton de la Fournaise volcano.

We have performed an additional survey in 1985 (100 stations) in the summit area. The three different data sets were combined together for an analysis.

Data processing

Bouguer anomalies are computed in the IGSN71 system and gravity values are tied to the base of Port des Galets (g=978917.40 mgal).

Topographic corrections have been computed with Hammer's tables and computer calculations, using digitized topography. For the marine survey, we used the latter method.

For the older data, the corrections performed at the BRGM were limited to 22 km. We added those from 22 to 120 km. For the summit zone data, a more detailed topographic grid allowed the use of computer calculations beyond 500 m.

The marine corrections computed for the 1980 map considered that the station is at sea

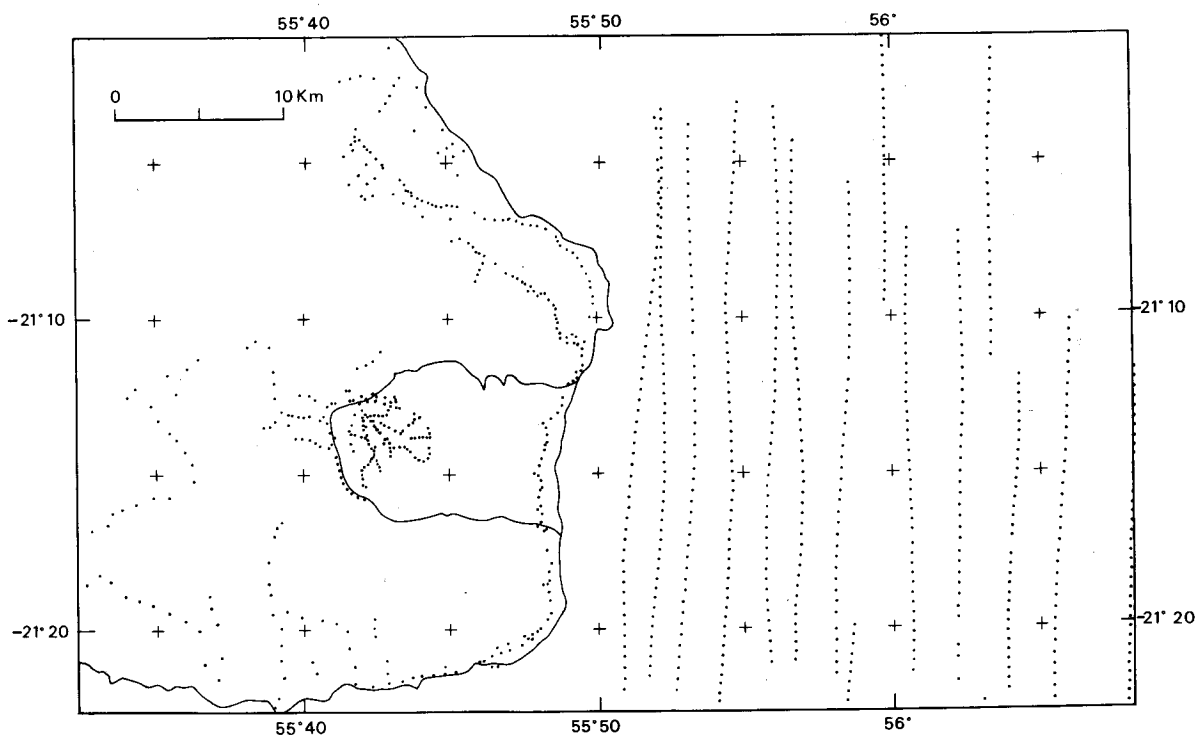

Fig. 2. Distribution of gravity stations.

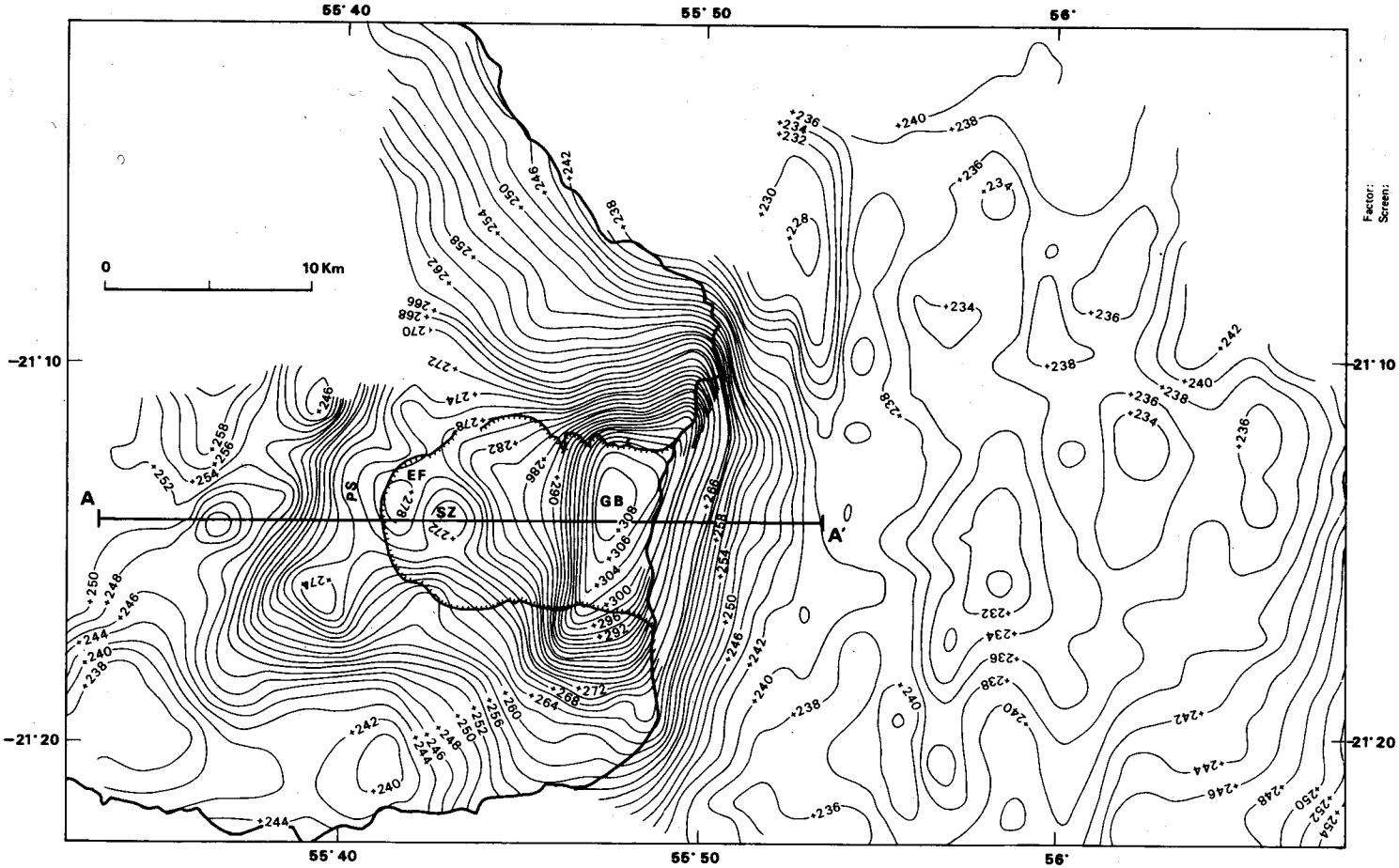

Fig. 3. Bouguer anomaly map for $d = 2.7$ g cm^{-3}. Contour interval is 2 mgal. *GB* = Grand Brûlé anomaly; *SZ* = summit zone anomaly; *PS* = Plaine des Sables anomaly; *EF* = Enclos Fouqué anomaly.

level. This approximation introduced a bias for the points of high elevation and near the shore, specially those of the summit zone, where the topographic corrections were underestimated. The very low gravity anomaly observed there in the former map (Gérard et al., 1980) must be partially regarded as an artifact. We have computed again the topographic corrections for the points of high elevation. For the low elevation points, we have just added the topographic correction from 22 to 120 km.

Choice of the density

The choice of the density used in Bouguer correction for a volcanic area is often discussed (e.g. Rymer and Brown, 1986). Of course, we do not know the exact density repartition and we must choose a homogeneous density as close as possible to the average density of the island. Although in their gravity studies over several volcanoes Williams and Finn (1985) in the Cascade Range and Kinoshita (1965) in Hawaii employ a low density ($d=2.2$, 2.3 g cm^{-3}) we have preferred after computing the Bouguer anomaly for 2.5, 2.6, 2.7 g cm^{-3} to use $d=2.7$ g cm^{-3} (Fig. 3) because of several considerations:

- The two cross-sections of the Bouguer anomaly (Fig. 4), show that the 2.7 g cm^{-3} seems to make the anomalies more distinct and the correlation between topography and Bouguer anomaly lower than with 2.5 g cm^{-3} density. However, because we are in an anomalous area, we cannot strictly use Nettleton's method.
- The 2.7 g cm^{-3} density is consistent with the density determined from the marine survey (Rousset et al., 1987).
- Last and perhaps the strongest argument we have some information on density evolution with depth in the two sites of Cirque de Salazie and Grand Brûlé (Fig. 1) studied by CFG and BRGM for geothermal exploration (Demange et al., 1984).

In Cirque de Salazie densities greater than 2.7 g cm^{-3} were determined for depths greater than 300 m, thus we can reasonably assume that the older and more compacted lavas of Piton des Neiges on which Piton de la Fournaise volcano lies have a mean density of 2.7 g cm^{-3}. On Piton de la Fournaise itself, in the Grand Brûlé borehole the denser basement is encountered below 800 m of low-densities lavas ($d=2.5$ g cm^{-3}). Even if these lavas are probably thicker in Piton de la Fournaise the inferred anomaly can be neglected compared to the contribution of the rest of the island.

Accuracy of the Bouguer anomaly

On-shore data

The accuracy of the anomaly depends on the height estimation. On the roads, the elevation is generally determined by geometric levelling and the accuracy is very good (some millimeters). Off the roads, in rough topography, the height is known only by the barometric method. On the summit zone a geometrical levelled network established after 1980 improved the reliability of barometric levelling during the 1985 survey and allowed a new elevation determination of the former survey stations. Barometric tide recording and numerous reoccupations yield an accuracy better than 2 m. Therefore, on-shore gravity anomalies (without topographic corrections) have an accuracy better than 0.4 mgal, if we assume a free-air Bouguer gradient of -0.1938 mgal/m. But topographic corrections, particularly the nearest ones, can be important and they are rather inaccurate. As a result, the final relative uncertainty between two neighbouring points can be assumed to be lower than 2 mgal.

Off-shore data

Owing to an imperfect acquisition, this set of data is not as accurate as it should be. The topographic corrections computed are smooth, and quite accurate. The accuracy is estimated to 2–3 mgal.

When we look at the map (Fig. 3) and the cross-section (Fig. 4) we can suspect that a

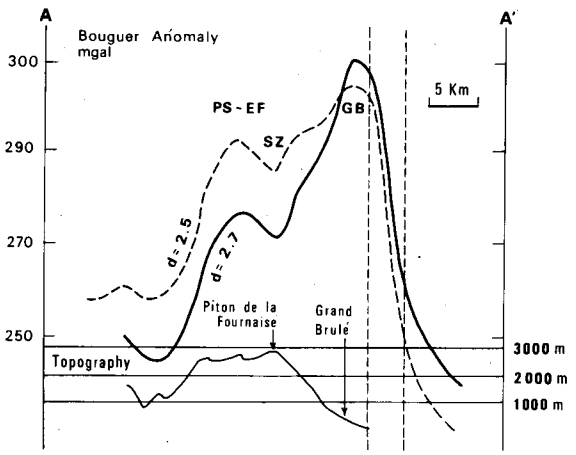

Fig. 4. Bouguer anomalies (computed with $d=2.5$ and $d=2.7$ g cm^{-3}) and topography profiles (see Fig. 3 for location). Vertical dashed lines indicate the zone without data. Bouguer and topographic corrections have opposed signs and their combination depend on density and elevation values. This explains that for elevations near sea level the Bouguer anomaly computed with $d=2.5$ g cm^{-3} becomes lower than the one computed with $d=2.7$ g cm^{-3}.

constant difference remains between the off- and on-shore data, probably due to a wrong gravity tie of the off-shore survey to the base of Port des Galets. This would explain the high gradient just at the boundary of the two zones, where we have no measurement to control the actual Bouguer value.

We will take this eventuality into account when posing the problem of the three-dimensional inversion of the residual gravity anomalies.

Bouguer anomaly map (Fig. 3)

Because of the difficulties of access in some areas, the data coverage is largely uneven. The contour map should therefore be studied carefully in the zones where the density of measured points is low. Moreover, the rough topography of this region can modify the shape of the anomalies even though topographic corrections have been carried out.

We see, at the center of the map the high anomaly of Grand Brûlé (GB, Fig. 4) which is closed near the shore. It is coalescent with a bigger anomaly, which involves most of the Fournaise area. This anomaly seems to end westwards between the limits of the two old calderas. But there, the Bouguer anomaly is still high, because of the positive axis as pointed out before by Gérard et al. (1980). In the summit zone, we can distinguish a positive anomaly in the western part of the younger caldera, the Enclos Fouqué (EF anomaly), and in the Plaine des Sables (PS anomaly).

A negative anomaly can be found in the south west part of the map, far from the centers of activity and almost certainly linked to superficial structures. More interesting is the negative anomaly associated with the summit cone of Piton de la Fournaise (SZ anomaly). Generally, basaltic volcanism is characterized (Rymer and Brown, 1986) by positive anomalies and for example, in Hawaii, the Bouguer anomaly map (Kinoshita, 1965) shows a regular increase from the shore to the highest zones, as is the case for the Canary islands (Mc Farlane and Ridley, 1968, 1969). This anomaly will be discussed with the interpretation of all the data and with the detailed gravity study of the summit zone.

In summary, the set of complex positive anomalies of Piton de la Fournaise volcano is clearly related to the N120° Rift zone and the ultramafic body of Grand Brûlé. But the NE- and SE-trending rift zones do not seem to be characterized by strong positive anomalies. The small positive anomalies found on the summit zone will be discussed separately.

Our studied area is small compared to the dimensions of the anomalies, so it is impossible to use analytical methods to separate the residual and regional field. The regional field will be determined according to the problem to be solved. For the global inversion of all the Fournaise data, we will use the simplest regional field we can find: a constant value. We can locate on the map (Fig. 3) and on the profile (Fig. 4), the regional level between 235 and 245 mgal which is consistent with the intra-oceanic setting of

the island. It will be introduced as a new unknown in the problem.

Three-dimensional inversion of the gravity data

Linear programming inversion

The aim of any gravity survey is to find out the distribution of the densities in the ground, using the disturbance produced in the Earth's gravity field. To interpret such phenomena there are two kinds of method: direct and inverse methods. To use a direct method one computes the gravity effect of a given structure, compare it to the data and, eventually corrects the structure to obtain a good fit between the model and the observations. On the contrary, inverse methods aim at extracting the information from the anomalies themselves. Of course, no way exists of determining realistic geological models with gravity measurements alone, the gravity problem itself has an infinite number of solutions. Therefore, we must place additional constraints on the problem: ground discretization, or statistical consideration of the possible densities, for example. This is a way of introducing a geological a priori in a numerical problem. After that, we shall look for the optimum solution. In linear programming we pose the problem as follows:

We have a set of M data (anomalies) d_i with an error e_i:

$$d_i - e_i < d_i < d_i + e_i \qquad 1 < i < M$$

We choose N unknowns p ($N \gg M$) bounded by:

$$p_{i\,\min} < p_i < p_{i\,\max} \qquad 1 < i < N$$

If the N unknowns are the densities of N elementary volumes, we have a linear undetermined problem to solve. We choose the optimum solution by minimizing a linear form of the unknowns of geological interest. This is a typical problem of linear programming which can be solved by the Simplex algorithm (Dantzig, 1963). Cuer and Bayer (1980a) have developed a complete software package devoted to the inversion of gravity (and magnetic) data. In linear programming, the optimization criterion does not assume any statistical hypothesis about the gravity signals and any model which satisfies the above inequations may be considered.

We see two main advantages in the use of this method to solve our problem:

- We have a poor knowledge of the inner structure of Piton de la Fournaise volcano, but we can find realistic geological bounds to the density contrasts.
- By choosing different linear forms to minimize, we always obtain extreme models which would constrain the geological interpretation of the studied area.

A basic theorem of linear programming states that for the solution which minimizes the linear form, among the $N+M$ upper inequations, N are strictly equations. This explains why most of the parameters are at their upper or lower bound.

Definition of the problem

Geometrical discretization

We divided the studied volume into 560 blocks of different size. The dimensions of the elementary volumes vary with their elevation and the coverage of the measurements above them. We discretized from 6000 m under sea level up to 2400 m above. The deepest blocks are 2000 m thick and the most superficial ones are 200 m thick. Bounds on the density contrast are 0.0 and 0.6 g cm^{-3}, generally and -0.2 and 0.0 g cm^{-3} under the negative anomaly of the southern coast.

Regional field

We can see, on the marine part of the map that the regional level is approximately 238 mgal. On-shore, as previously noted a difference may remain between the on- and off-shore surveys. So, we introduce two supplementary unknowns which correspond to two different

offsets of the gravity anomalies. We consider the regional value as constant on the zone and limited by 235 and 245 mgal. The problem is to find out the set of 562 values of ρ which satisfies:

$$\begin{pmatrix} d_1 - e_1 \\ '' \\ '' \\ d_{M-1} - e_{M-1} \\ d_M - e_M \end{pmatrix} \leqslant \begin{pmatrix} g_1 \cdots & g_{1,N-1} & g_{1,N} \\ '' & '' & '' \\ '' & '' & '' \\ g_{M-1,1} & g_{M-1,N-1} & g_{M-1,N} \\ g_{M,1} & g_{M,N-1} & g_{M,N} \end{pmatrix} \begin{pmatrix} \rho_1 \\ '' \\ '' \\ \rho_{N-1} \\ \rho_N \end{pmatrix} \leqslant \begin{pmatrix} d_1 + e_1 \\ '' \\ '' \\ d_{N-1} + e_{M-1} \\ d_M + e_M \end{pmatrix}$$

and minimizes a linear form of the unknowns.

$g_{i,N-1} = 1$ and $g_{i,N} = 0$ if the i value refers to off-shore data, $g_{i,N-1} = 0$ and $g_{i,N} = 1$ for on-shore data. The last unknowns ρ_{N-1} and ρ_N are respectively the offsets of the off- and on-shore data.

Data

This kind of program is limited for the number of data to approximately 250 (Cuer and Bayer, 1980b). We sampled 152 points as regularly distributed as possible on shore and belonging to the nearest tracks from the coast, for the off-shore survey. The error attributed to the data must take into account the experimental error itself and the approximation of the discretization, compared to the actual topography. The minimum uniform error which allowed us to find a first solution (necessary for the search of the desired solutions) is 4 mgal. This value is approximately twice the assumed error on the gravity anomaly.

Search for the particular solutions

We solved two elementary problems: the minimization of the total mass above a certain height, and the determination of the ideal body of Parker (Parker, 1975).

Minimization of the total mass. We want to minimize the anomalous mass above a certain level to determine if superficial sources are needed to explain the anomalies. So, we chose the total mass above two levels as the linear forms to minimize.

Above sea level. We can see on Figs. 5a and 6a the result of this computation. It is not possible to keep excess mass below sea level. Some small prisms must retain a high density. This is necessary to fit the anomalies of the western part of the Enclos Fouqué and of the Plaine des Sables caldera. Therefore there is a big gap between the few dense prisms which are as high as possible to minimize the mass and the other prisms, under the minimized volume. At depths, the dense body of Grand Brûlé lies just under sea level and vanishes in the last layer, between 4000 and 6000 m deep.

Above −1000 m. When minimizing from −1000 m to the top of the volcano (Fig. 6b), more mass remains in the highest layers. This fact could appear a little paradoxical but it can be clearly understood when looking at the −1000 m to sea level layer. In this layer, most of the mass has disappeared, except on the eastern coast. Consequently, the anomalies which needed a lot of mass in the −1000 m to sea level layer to be explained are now caused by less, but shallower mass. The presence of the mass, in spite of the minimization criterion, in the −1000 m to sea level layer, is in good agreement with the results of the Grand Brûlé geothermal drill hole, where the first massive compacted rocks ($d = 2.9$ at least) are found above −1000 m.

Conclusion. In both minimizations, we observe a conspicuous mass deficit under the summit zone. As a test, we tried to fill this zone with a homogeneous density. It appeared possible to use densities varying from −0.2 to 0.3 g cm^{-3}.

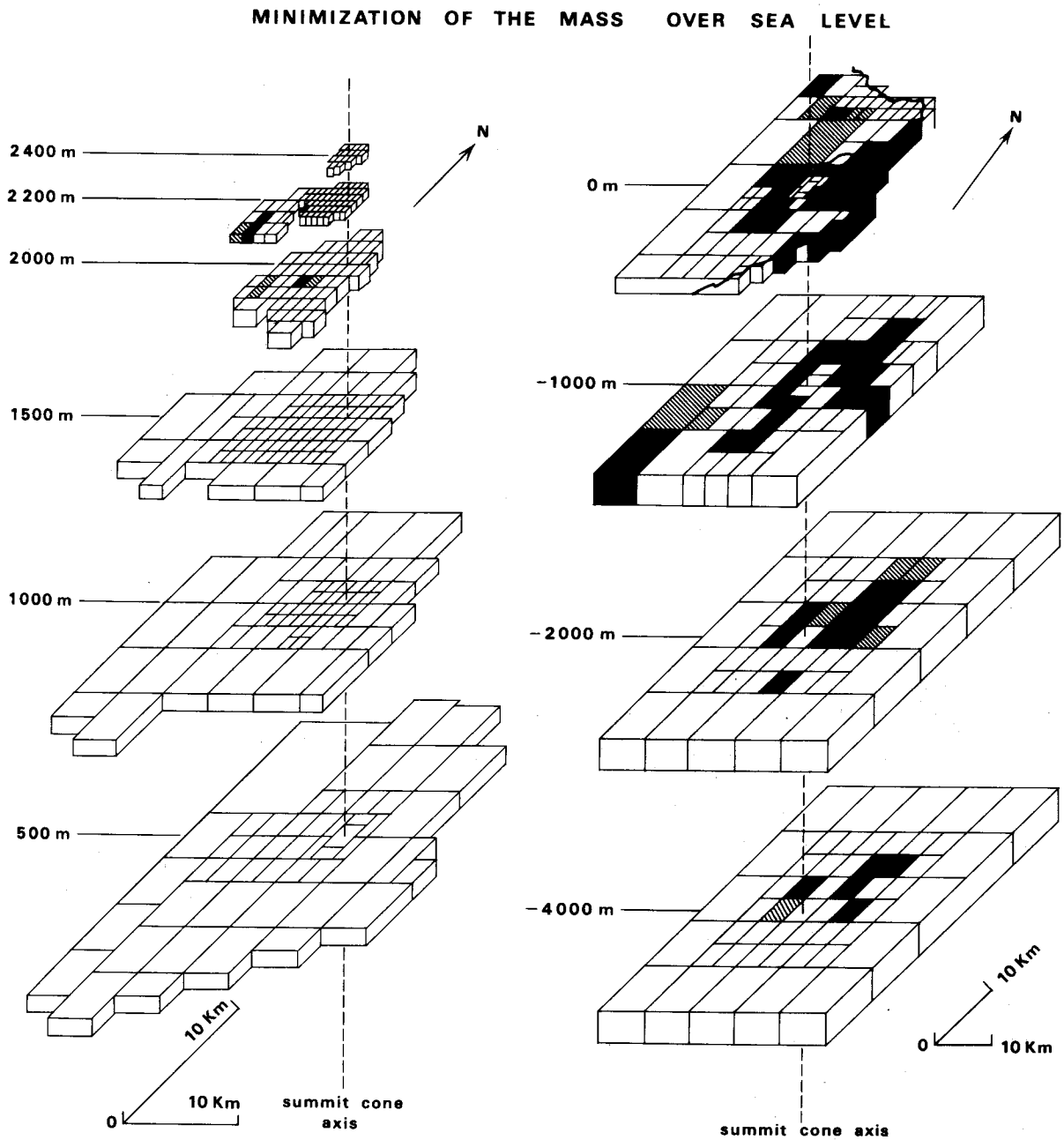

Fig. 5. Three-dimensional inversion using linear programming. The studied domain is separated in 10 layers; each layer is composed of prisms in which the density contrast can vary between 0.0 and 0.6 g cm^{-3}.
a. Computed model which minimizes the mass over sea level. Black prisms $d=0.6$ g cm^{-3}; white prisms $d=0.0$ g cm^{-3}; shaded prisms $d=$ intermediate value.

This means that we have certainly a mass difference between the central active zone of Piton de la Fournaise volcano and the surrounding rocks whose absolute value cannot be easily estimated.

These two models suggest the presence of

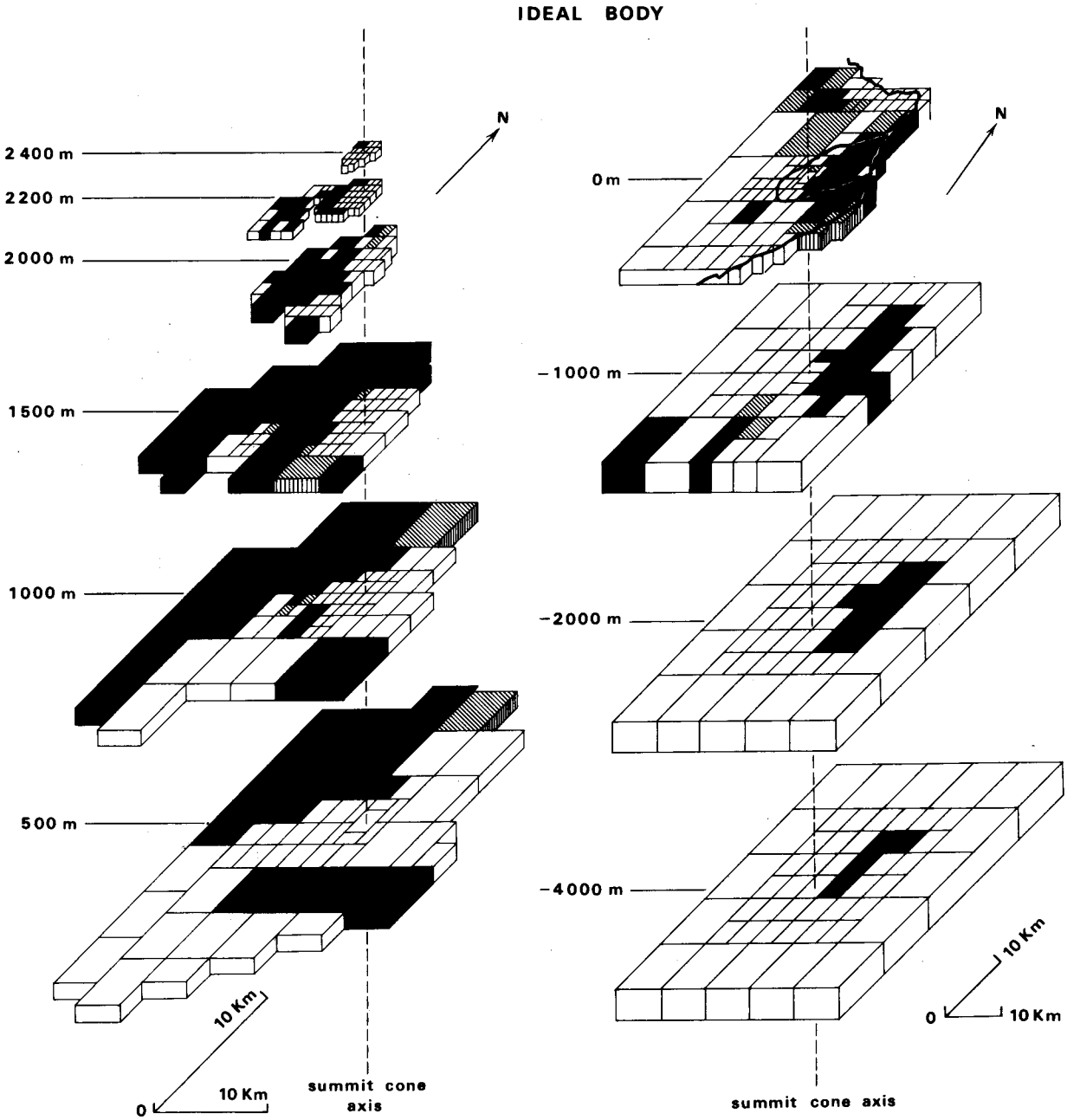

Fig. 5. b. Ideal body solution. Black prisms ideal body density ($d = 0.56$ g cm^{-3}); white prisms $d = 0.0$ g cm^{-3}; shaded prisms d = intermediate value.

dense rocks at intermediate depth on the West of the central zone of Piton de la Fournaise. Such rocks could be related to the activity of the N120° rift zone which would be similar to Kilauea East Rift Zone (Broyles et al., 1979).

Ideal body. Parker's ideal body solution is a model which minimizes the maximum value of

the density. Consequently, it is a homogeneous model and allows more excess mass high in the topography. In our case, it is not a great problem because the Grand Brûlé anomaly constrains the solution to keep a high density contrast as the minimum value. Therefore, the density contrast of the ideal body is 0.56 g cm^{-3} which is very close to its maximum value (0.6 g cm^{-3}) and to the expected values of the ultramafic rocks evidenced in the Grand Brûlé drill hole.

In this model, we can observe (Figs. 5b and 6c) the massive ultramafic body of Grand Brûlé, heavy rocks under the Plaine des Sables caldera and in the northern and southern flank of the volcano:

- Because it is well defined by steep gradients and a high anomaly, the Grand Brûlé intrusion is not really affected by the different choices of the linear form to be minimized and this solution is similar to the solution obtained by mass minimization.
- On the west of the central zone of the Piton de la Fournaise volcano, under the Plaine des Sables and Rivière des Remparts calderas, the model looks very different from those deduced from the mass minimizations (Fig. 6). We can distinguish a high-density layer under the surface, in the Enclos which is approximately 1500 m thick. It is not linked to any deep dense zone and seems to be completely disconnected from the ultramafic intrusion. However, we must remember that the ideal body solution is an extreme solution. The presence of mass excess in northern and southern flanks could be related to the activity of the NE- and SE-trending rift zones (Kieffer et al., 1977). This is very interesting but not very well constrained. Since there are not many measurements on these zones, it is easier to make the anomalous mass higher. Although positive anomalies do not appear in this zone, our data are consistent with such intrusive rift zones.

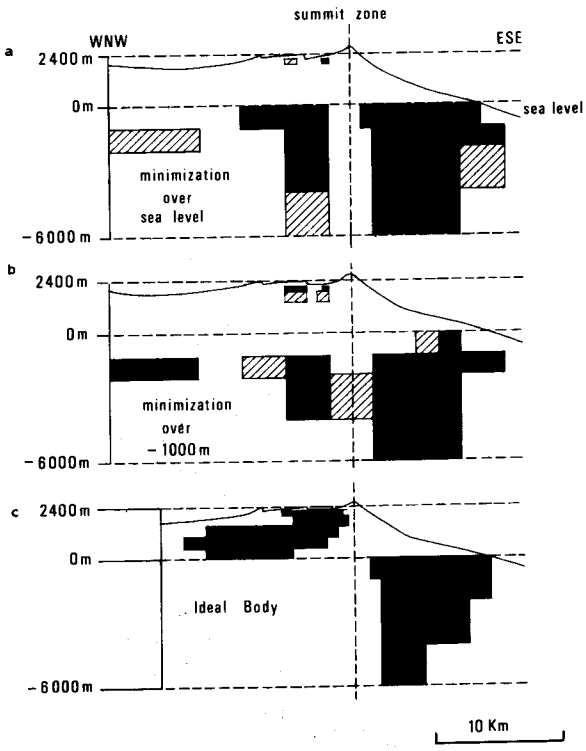

Fig. 6. Three-dimensional inversion using linear programming. WNW-ESE sections across the three computed solutions. Black prisms $d=0.6$ g cm^{-3}; shaded prisms $0.0<d<0.6$ g cm^{-3}. a. Minimization over sea level. b. Minimization over -1000 m. c. Ideal body.

Detailed gravity study of the summit zone

Because of the errors we had to accept in the three-dimensional inversion, the sampling of the data, and the discretization, it was not possible to take into account the small wavelength and amplitude anomalies of the summit zone during the study of the whole Fournaise area by inverse method.

Figure 7 shows the detailed map of the summit zone of Piton de la Fournaise volcano. The negative SZ anomaly (under 270 mgal) associated with the central cone is separated into two parts by a sharp positive anomaly in the West of Dolomieu crater. The positive EF anomaly

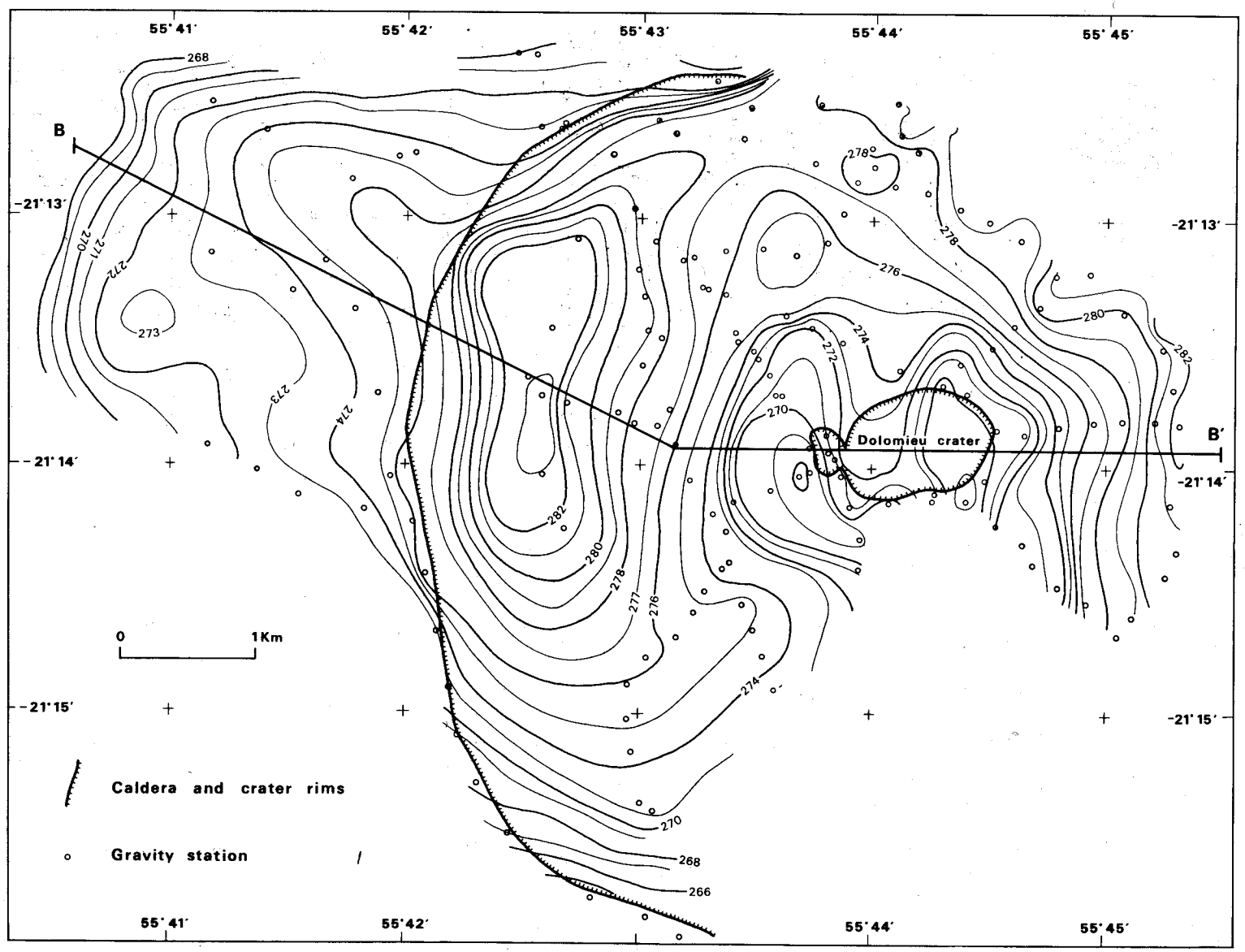

Fig. 7. Bouguer anomaly map ($d = 2.7$ g cm^{-3}) of the summit zone. Contour interval 1 mgal.

on the west of the Enclos Fouqué extends under the Plaine des Sables caldera (PS anomaly) before ending near the limit of the Rivière des Remparts caldera, as previously noticed. In some places around the summit cone, we can see some small poorly defined anomalies.

The negative anomaly of the summit zone

Since the negative anomaly is not modified much when the 2.5 g cm^{-3} density is used to compute the Bouguer anomaly, it is clear that this anomaly is not an artifact caused by a bad density used to compute the terrain corrections.

With the inversion of the gravity data, we studied the eventuality of a light column under the summit cone. Such very low-density rocks have been seen in the Kilauea volcano (Keller et al., 1979) for some hundreds of meters, but to create the SZ anomaly, we need light rocks over several kilometers. Therefore, the negative anomaly must be regarded chiefly as a relative minimum between two strong and coalescent positive anomalies (PS-EF and GB anomalies, Fig. 8), caused by massive dense rocks. This minimum is probably enhanced by the negative effect of superficial lavas and pyroclastic rocks of the summit area.

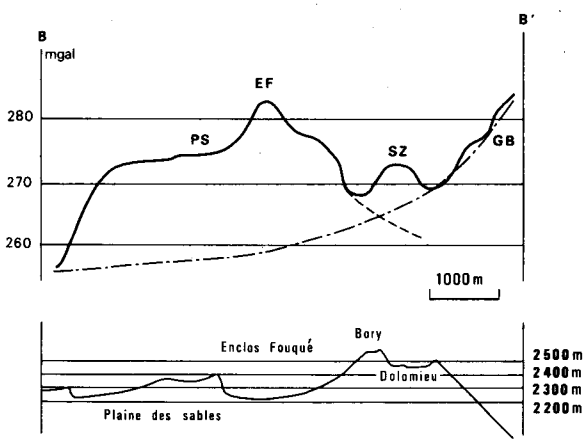

Fig. 8. *BB'* topographic (lower part of the figure) and Bouguer anomaly (upper part) profiles across the summit zone (see Fig. 7 for location). Solid line = Bouguer anomaly; dashed lines = extrapolation of the effect of the Grand Brûlé and Plaine des Sables heavy zones.

The sharp positive anomaly of the summit zone

This positive anomaly is not well defined. The steep gradients (Fig. 9) mean that the anomaly is related to a superficial and high-density structure. One solution, in terms of a homogeneous model, was determined by direct three-dimensional computation. The density contrast which allows a good fit between the computed and observed anomaly must be higher than 0.3 g cm^{-3}. The computed model for 0.3 g cm^{-3} density contrast is shown in Fig. 10. The intrusive zone, 700 m wide under the summit craters, becomes narrower at depth. The actual bottom of such a narrowing structure cannot be accurately determined. Northwards, it also becomes thinner.

Unfortunately, we have only a few stations on the southern flank of the cone but we have some indication of an extension of the intrusive zone there. This structure is certainly related to the present activity of Piton de la Fournaise volcano and we interpret it as a dyke zone. The shape of the anomaly can be correlated with a

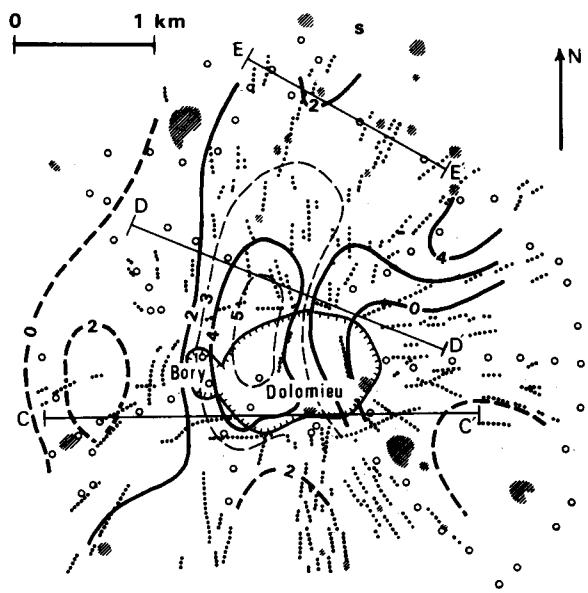

Fig. 9. Residual anomaly map of the summit zone and relations with active structures. Open circles = gravity stations; shaded area = emissive cone; dotted line = emissive fissure. Contour interval 1 mgal.

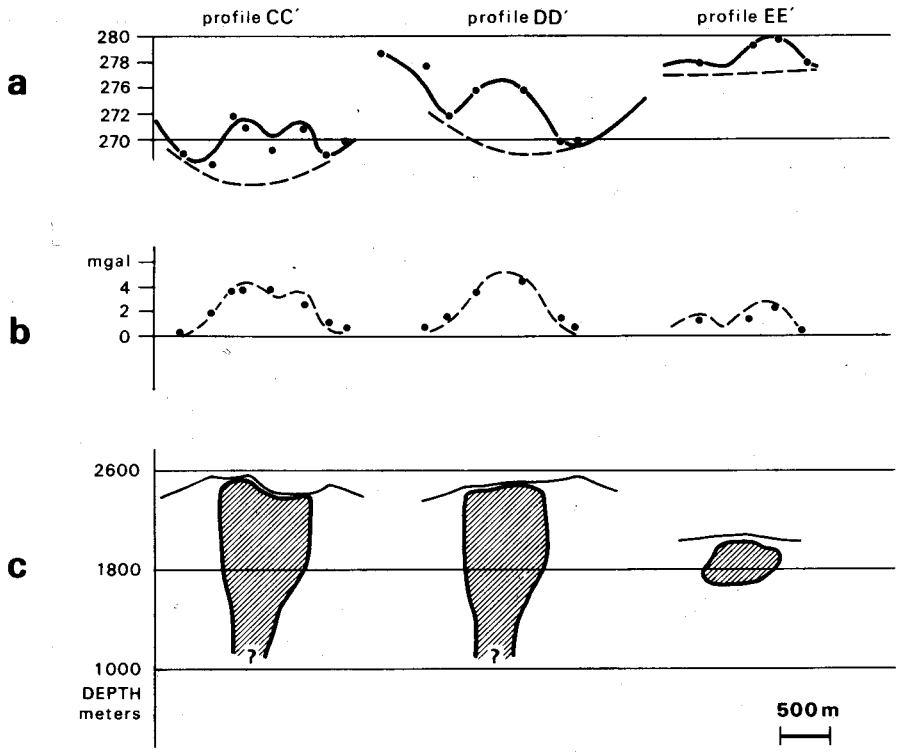

Fig. 10. Direct three-dimensional study of the summit zone. The computed model is shown on three sections (see Fig. 9 for location). a. Bouguer anomaly profile ($d = 2.7$ g cm^{-3}). Dashed line = regional anomaly used; solid line = smoothed Bouguer anomaly; dots = actual nearest values. b. Computed anomaly (dots) and residual anomaly (dashed line). c. Cross sections through three-dimensional model for a homogeneous density of 0.3 g cm^{-3}.

preferential intrusive zone (Fig. 9) corresponding with the intersection of the two NE- and SE-trending rift zones. However, we must note that geologic studies do not reveal a sufficient amount of intrusive rocks to explain such a density contrast. Therefore, we suspect the presence of more massive lavas at depths.

Volcanic seismicity locates the top of a shallow magma chamber at 1000 m under the summit craters i.e. in the lower part of the intrusive body (Lénat et al., 1989).

Conclusion

The cross section (Fig. 8) suggests that the long wavelength anomaly of the Plaine des Sables caldera is of deep or intermediate depth origin as already discussed. Under the Enclos Fouqué, the origin of the EF anomaly is more superficial. These structures could be related to the former eruptive stages of the Piton de la Fournaise volcano whose center of activity migrated from the west of the Enclos Fouqué to its present location (Chevallier and Bachèlery, 1981).

Various positive anomalies are evidenced by small inflexions of the contour curves. Not all of them are well defined but we can see (Fig. 9) a good correlation between the residual anomaly (contours) and the active fissures.

Summary and discussion

In all the solutions, the geometry of the ultramafic body of Grand Brûlé appears approximately constant. The high gradients on the north, the south and the east clearly show its limits. The very high amplitude of the anomaly

corresponds to a density of 3.2–3.3 g cm^{-3} and a thickness of 5–6 km. This density corresponds to the value determined in the drill-hole and thus confirms the good choice of the density Bouguer correction.

This body is completely cooled and can be considered as an old layered magma chamber (Demange et al., 1984). Although we have not yet age data, we can assume an age older than 500.000 years for this body, on the basis of a simple cooling model and magnetic anomalies considerations (Rançon et al., 1987). For this reason, at least, it is not possible to associate this magma storage structure with the feeding system of the present volcanism of Piton de la Fournaise volcano. Earlier geological studies (Chevallier and Bachèlery, 1981) give an age of 400,000 years for the first stage of Piton de la Fournaise volcano. We have to assume the presence of an older volcano of which no obvious aerial structure remains.

The anomalous topographic feature, east of Grand Brûlé (Rousset et al., 1987) could be related to the "volcano of Grand Brûlé". It is clear that this body is the major structure on the eastern flank of Piton de la Fournaise volcano and it is certainly an important guideline for the structural evolution of this zone. For example, it could be partly responsible for the collapse of the eastern part of the Enclos Fouqué caldera. We note the presence of a similar anomaly in the center of Piton des Neiges volcano (Gérard et al., 1980), associated with ultramafic rocks outcropping in Cirque de Salazie.

To explain the PS and EF anomalies, we propose an intermediate solution between the two following extreme ones. Either the intrusive rocks are situated at great depth and perhaps in continuity with the Grand Brûlé intrusion, along the N120 rift zone, or the dense structures bounded by caldera rims are superficial and belong to the edifice of Piton de la Fournaise volcano.

Whatever solution we choose, dense superficial rocks must exist. These superficial intrusions, in Plaine des Sables and Enclos Fouqué calderas are interpreted, like the anomaly of the summit zone (see above), as dyke complexes associated with the intrusive systems which crop out in Plaine des Sables and Calderas rims. We interpret the large PS anomaly as a deeper (over 2 km below sea level) magma storage structure which fed the shallow dyke systems active during the former stages of Piton de la Fournaise volcano evolution. We propose that the ultramafic xenoliths found in some lavas in this area (Bachèlery, 1981) are surface evidence of these accumulation structures. In the central zone, we infer that the lack of mass is related to the active feeding conduit which pierces the older lavas.

This work leads to a more complex structure of Piton de la Fournaise volcano, as well as a more complex history of the whole island, than has been presented before. For example, the hypothesis of a northwest to southeast migration (e.g. Chevallier and Bachèlery, 1981) of the volcanism seems to be no longer valid.

Acknowledgements

This work, including the new gravity survey, has been supported by the P.I.R.P.S.E.V. (volcanological program of the Centre National de la Recherche Scientifique). We must thank all the people who have worked to perform gravity measurements on Piton de la Fournaise during the last years; and the BRGM and the ORSTOM which provide us with data.

References

Bachèlery, P., 1981. Le Piton de la Fournaise (Ile de la Réunion). Etude volcanologique, structurale et pétrologique. Thesis, University of Clermont-Ferrand, 259 pp.

Broyles, M.L., Suyenaga, W. and Furumoto, A.S., 1979. Structure of the lower East Rift Zone of Kilauea volcano, Hawaii, from seismic and gravity data. J. Volcanol. Geotherm. Res., 5: 317–336.

Chevallier, L. and Bachèlery, P., 1981. Evolution structurale du volcan actif du Piton de la Fournaise, Ile de la Réunion, Océan Indien Occidental. Bull. Volcanol., 44(4): 723–741.

Chevallier, L. and Vatin-Pérignon, N., 1982. Volcano struc-

tural evolution of Piton des Neiges, Réunion Island, Indian Ocean. Bull. Volcanol., 45(4): 285-298.

Cuer, M. and Bayer, R., 1980a. Fortran routines for linear inverse problems. Geophysics, 45: 1706-1709.

Cuer, M. and Bayer, R., 1980b. A package of routines for linear inverse problems (user and technical manual). Cah. Math., 18, Univ. Sci. Techn. du Languedoc, Montpellier.

Dantzig, G.B., 1963. Linear Programming and Extensions. Princeton University Press, Princeton, NJ.

Demange, J., Pouilland, N.N. and Rançon, J.P., 1984. Rapport CFG-BRGM; 84GTH001, 51 pp.

Duffield, W.A., Stieltjes, L. and Varet, J., 1982. Huge landslides blocks in the growth of Piton de la Fournaise, la Réunion, and Kilauea volcano Hawaii. J. Volcanol. Geotherm. Res., 12: 147-160.

Gérard, A., Lesquer, A., Lachaud, J.C., Louis, P. and Mennechet, C., 1980. Etude gravimétrique de la moitié Sud Est de l'île de la Réunion. C.R. Acad. Sci. Paris, Sér. B, 290: 139-142.

Keller, G.V., Grove, L.T., Murray, J.C. and Skokan, C.K., 1979. Results of an experimental drill hole at the summit of Kilauea volcano, Hawaii. J. Volcanol. Geotherm. Res., 5: 345-385.

Kieffer, G., Tricot, B. and Vincent, P.M., 1977. Une éruption inhabituelle (Avril 1977) du Piton de la Fournaise (Ile de la Réunion): ses enseignements volcanologiques et structuraux. C.R. Acad. Sci. Paris, Sér. D, 285: 957-960.

Kinoshita, W.T., 1965. A gravity survey of the island of Hawaii. Pac. Sci., 19: 339-340.

Lénat, J.F. and Aubert, M., 1982. Structure of Piton de la Fournaise volcano (La Réunion Island, Indian Ocean) from magnetic investigation. An illustration of the analysis of magnetic data in a volcanic area. J. Volcanol. Geotherm. Res., 12: 361-392.

Lénat, J.F., Bachèlery, P., Bonneville, A., Tarits, P., Cheminée, J.L. and Delorme, H., 1989. The December 1983-January 1984 eruption of Piton de la Fournaise volcano, Réunion Island. J. Volcanol. Geotherm. Res., 36: 87-112.

Lecointre, J. and Vincent, P.M., 1984. Séquence de pyroclastites différenciées sur le massif du Piton des Neiges. Ile de la Réunion. Caractéristiques et signification volcano-structurale. Abstract in Colloque Pirpsev, Clermont Ferrand.

McDougall, I., 1971. The geochronology of the young volcanic island of Réunion, Indian Ocean. Geochim. Cosmochim. Acta, 36: 261-288.

McFarlane, D.S. and Ridley, W.I., 1968. An interpretation of gravity data for Tenerife, Canary Islands. Earth Planet. Sci. Lett., 4: 481-486.

McFarlane, D.S. and Ridley, W.I., 1969. An interpretation of gravity data for Lanzarote Canary Islands. Earth Planet. Sci. Lett., 6: 431-436.

Parker, R.L., 1975. The theory of ideal bodies for gravity interpretation. Geophys. J. R. Astron. Soc., 42: 315-334.

Rançon, J.P., Lerebour, P. and Auge, T., 1987. Mise en évidence par forage d'une chambre magmatique ancienne à l'aplomb de la zone orientale du Piton de la Fournaise (île de la Réunion): implications volcanologiques. C.R. Acad. Sci. Paris, 304(II): 55-60.

Rousset, D., Bonneville, A. and Lénat, J.F., 1987. Detailed gravity study of the off-shore structure of Piton de la Fournaise Volcano, Réunion Island. Bull. Volcanol., 49: 713-722.

Rymer, H. and Brown, G.C., 1986. Gravity field and the interpretation of volcanic structures: Geological discrimination and temporal evolution. J. Volcanol. Geotherm. Res., 27: 229-254.

Williams, D.L. and Finn, D., 1985. Analysis of gravity data in volcanic terrain and gravity anomalies and subvolcanic intrusions in the Cascade Range, U.S.A., and at other selected volcanoes. In: J. Hinze (Editor), The Utility of Regional Gravity and Magnetic Anomaly Maps. Soc. Explor. Geophysicists, Tulsa, Okl.